YUWEI SUN

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EDUCATION

The University of Tokyo — Tokyo, Japan
Ph.D., Information & Communication Engineering — 04-2021 ~ Present
M.E., Information & Communication Engineering — 04-2019 ~ 03-2021
GPA: 3.84/4.0
Research Focus: Deep Learning, Cybersecurity, Trustworthy AI

Post-Graduate Research Program, Graduate School of Information Science & Technology — 10-2018 ~ 03-2019
Research Focus: Deep Learning, Cybersecurity

North China Electric Power University — Hebei, China
B.E., Computer Science and Technology — 09-2014 ~ 09-2018
Thesis: An Attack on Deep Learning Systems Based on Generative Adversarial Networks

ADDITIONAL EDUCATION

Massachusetts Institute of Technology — Cambridge, MA, US
Fellow of the Advanced Study Program, Graduate School of Engineering — 02-2020 ~ 05-2020
Courses: Emergent Computations within Distributed Neural Circuits, Underactuated Robotics, Blockchain Lab
Research Focus: Neural Networks, Dynamic Systems

University of Pennsylvania — Philadelphia, PA, US
English Language Program — 08-2019 ~ 10-2019

Chulalongkorn University — Bangkok, Thailand
Visiting Student, Department of Mathematics and Computer Science — 02-2019 ~ 03-2019
Research Focus: Computer Vision, Object Recognition

Waseda University — Tokyo, Japan
Japanese Language Program — 10-2016 ~ 08-2017

EMPLOYMENT

RIKEN Center for Advanced Intelligence Project — Tokyo, Japan
Junior Research Associate, AI Security and Privacy Team — 04-2021 ~ Present
The Center has been launched for the Advanced Integrated Intelligence Platform Project (AIP) of the Ministry of Education, Culture, Sports, Science and Technology (MEXT).
- *Research on AI Security and Privacy for World Model Sharing*

United Nations University Centre — Tokyo, Japan
Systems Engineer Intern, Campus Computing Centre (C3) — 05-2019 ~ 12-2020
C3 manages the University's information and communication technology resources and network infrastructure.
- *Supported work of C3 through network systems data analysis and research*
- *Worked with team members on the research project and report at meetings of the ICT department*
- *Researched on a collaborative deep learning platform for data sharing with privacy considered*

AIESEC Global Talent Program (Value Bridge Company) — Tokyo, Japan
Software Engineer, ICT Department — 03-2018 ~ 09-2018
- *Contributed to developing and operating with automatic product quality inspection systems*
- *Developed cloud photo recommendation systems with facial recognition, which extended business of the company*

PUBLICATIONS

Yuwei Sun, Hideya Ochiai, Hiroshi Esaki. Adaptive Intrusion Detection in The Networking of Large-Scale LANs with Segmented Federated Learning. *IEEE Open Journal of the Communications Society, Vol.2, pp.102-112. 2020.* (peer reviewed)

Yuwei Sun, Hideya Ochiai, Hiroshi Esaki. Intrusion Measurement and Detection in LAN Using Protocol-Wise Associative Memory. *IEEE International Conference on AI in information and communication (ICAIIC). 2021*. (oral presentation, peer reviewed)

Yuwei Sun, Hideya Ochiai, Hiroshi Esaki. Deep Learning-Based Anomaly Detection in LAN from Raw Network Traffic Measurement. *IEEE International Conference on Information Sciences and Systems (CISS). 2021.* (oral presentation, peer reviewed)

Yuwei Sun, Ng S.T. Chong, Hideya Ochiai. Network Flows-Based Malware Detection Using a Combined Approach of Crawling and Deep Learning. *IEEE International Conference on Communications (ICC): Next-Generation Networking and Internet Symposium. 2021.* (oral presentation, peer reviewed)

Yuwei Sun, Hideya Ochiai, Hiroshi Esaki. Multi-Type Malware Detection in A Real-World Network Using Raw Network Traffic. *IEEE Consumer Communications & Networking Conference (CCNC). 2021.* (poster, peer reviewed)

Yuwei Sun, Hideya Ochiai, Hiroshi Esaki. Blockchain-Based Federated Learning Against End-Point Adversarial Data Corruption. *IEEE International Conference on Machine Learning and Applications (ICMLA). 2020.* (oral presentation, peer reviewed)

Yuwei Sun and Russ Tedrake. Trajectory Optimization for An Autonomous Vehicle Driving across Stochastic Traffic Flows based on Direct Collocation. *IEEE International Conference on Control, Automation and Diagnosis (ICCAD). 2020.* (oral presentation, peer reviewed)

Yuwei Sun, Hideya Ochiai, Hiroshi Esaki. Visual Analytics for Anomaly Classification in LAN Based on Deep Convolutional Neural Network. *IEEE International Conference on Informatics, Electronics and Vision (ICIEV). 2020.* (oral presentation, peer reviewed)

Yuwei Sun, Hideya Ochiai, Hiroshi Esaki. Intrusion Detection with Segmented Federated Learning for Large-Scale Multiple LANs. *IEEE International Joint Conference on Neural Networks (*IJCNN*). 2020.* (oral presentation, peer reviewed)

Yuwei Sun, Ng S. T. Chong, Hideya Ochiai. Text-based Malicious Domain Names Detection Based on Variational Autoencoder And Supervised Learning. *IEEE International Conference on Information Sciences and Systems (CISS). 2020.* (oral presentation, peer reviewed)

Yuwei Sun, Hideya Ochiai, Hiroshi Esaki. Detection and Classification of Network Events in LAN Using CNN. *IEEE International Conference on International Conference on Information Technology (InCIT). 2019.* (oral presentation, peer reviewed)

Yuwei Sun, Nagul Cooharojananone, Hideya Ochiai. Aircraft Detection Based on Saliency Map and Convolution Neural Network. *IEEE International Conference on International Computer Science and Engineering Conference (ICSEC). 2019.* (oral presentation, peer reviewed)

INVITED TALKS

- Workshop on Algorithm and Big Data, Transdisciplinary Information Sciences Conferences. Online 03-2021
- UniNet Workshop, Thailand Ministry of Higher Education. Thailand 01-2020

HONORS AND AWARDS

- The University of Tokyo, Department Chair's Award 03-2021
- The University of Tokyo, Fellow of the UTokyo International Student Scholarship 10-2019
- International Association of Students in Economic and Commercial Sciences (AIESEC), Leadership Development Experience Certificate 09-2018

- North China Electric Power University, Fellow of the Excellent Student Scholarship 12-2016
- Consortium for Mathematics and Its Applications (COMAP), Mathematical Contest In Modeling, Successful Participant 12-2015

SKILLS

Programming: Python (Advanced), Tensorflow (Intermediate), OpenCV (Intermediate), SQL (Intermediate), Linux commands (Intermediate), C++ (Intermediate), PyTorch (Elementary), Java (Elementary), HTML (Elementary), Git (Elementary), JavaScript (Elementary)
Platform: Google Cloud Platform (GCP) (Elementary), Amazon Web Services (AWS) (Elementary)
Languages: Chinese (native), English (TOEFL IBT 101), Japanese (N1 169)

CERTIFICATION

- United Nations Office for Disarmament Affairs, AI Governance Workshop Certificate 02-2021
- International Innovation Center of Tsinghua University, AUA Entrepreneurship Initiative Online Program Certificate 12-2020
- Japan Deep Learning Association (JDLA), Deep Learning for General Certification 11-2018
- Japan Deep Learning Association (JDLA), Deep Learning for Engineer Certification 09-2018